

Research on Seismic Signal Classification and Recognition Based on EEMD and CNN

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Abstract—The accurate classification of seismic signals is of great significance for updating clear earthquake catalog, early-warning the arrival of strong earthquake, and further researches in the field of seismology. This paper first preprocesses the seismic signal data of two types of events: earthquakes and explosions, and then uses the ensemble empirical mode decomposition (EEMD) method to decompose these preprocessed seismic data. A set of intrinsic mode functions being distributed from high-frequency to low-frequency are obtained from each seismic waveform, and then corresponding Hilbert spectrum is obtained by applying Hilbert transform to each of these intrinsic mode functions (IMFs). The image of Hilbert spectrum is scaled to a suitable and comparable size, by repeated trial-and-error experiments, to the size of 32*32 pixels, and then the 32*32 pixels gray-scale image is fed to a convolutional neural network(CNN) to classify the types of seismic signals. In this paper, 1674 waveforms of 54 earthquake events and 1509 waveforms of 63 explosion events are processed as above procedures. By five-fold cross-validation, 80% of the data is used as the training set, remaining 20% data as testing set. The highest classification accuracy rate of testing set is 94.98% with the average accuracy rate 94.41%, this results are significant higher than those of multi-layer perceptron(MLP) and Support Vector Machine(SVM), which implying stronger classification capability of the scheme unifying EEMD and CNN proposed in this paper.

Keywords—Seismic Signals Classification; EEMD; CNN; Earthquake; Explosion

I. INTRODUCTION

The classification of seismic signals is still a hot topic in the field of seismology. With the continuous improvement of the accuracy of seismic station instruments, we have collected and recorded more and more seismic signals, and more and more types. The largest number of seismic signals collected by seismic stations are mainly tectonic earthquakes in natural earthquake and explosion in unnatural earthquake. These unclassified seismic signals will contaminate the earthquake catalog, affect further research in the field of seismology, and have a significant impact on the early warning the arrival of strong earthquake. Therefore, the accurate classification of different types of seismic signals is of great significance for the updating clear of earthquake catalogs, further research in the field of seismology, and real-

time warning of strong earthquakes. At present, these tasks are still completed by inefficient manual [1].

In order to solve the problem of seismic signal classification, many researchers have developed a variety of methods. In the works of [2-6], the characteristics of the seismic signal's body wave magnitude to surface wave magnitude ratio, waveform complexity, spectrum ratio, angular frequency and seismic distance are extracted, and use support vector machine (SVM), principal component analysis (PCA) and multivariate statistical analysis to classify different types of seismic signals. In the works of [7-14], starting from the acoustic signal processing method, extracting the acoustic characteristics of seismic signals, using continuous hidden Markov models, discrete hidden Markov models and statistical hidden Markov models to classify seismic signals. In the works of [15-20], the neural network method is used to identify, classify and locate seismic signals.

The time-frequency analysis method is an important method of seismic signal analysis. The ensemble empirical mode decomposition(EEMD) method is an important method of the signal time-frequency analysis method. It is an adaptive decomposition method suitable for the analysis of unsteady and nonlinear seismic signals. It is based on the empirical identification of the characteristic time scale of the vibration mode in the data, and then the data is decomposed accordingly. Its decomposition result can smooth the non-stationary signal and make the instantaneous frequency meaningful. In the works of [21-23], used empirical mode decomposition(EMD) or EEMD to classify seismic signals and achieved good classification results.

In our works, we first perform the necessary data preprocessing on the original seismic signal, and then use the EEMD method to decompose the seismic signal to obtain a set of intrinsic mode function(IMF) components from high to low frequencies, and then find the instantaneous frequency and amplitude of each IMF component, and Use the Hilbert transform to transform each IMF component to obtain a three-dimensional Hilbert spectrum containing time, frequency and amplitude, and construct a convolutional neural network model, the generated Hilbert spectrum is used as the input of neural network to classify natural earthquake and explosion signals. Figure 1 shows the flow chart of the method in this paper. Experimental results show that the method in this paper can effectively classify earthquakes and explosions.

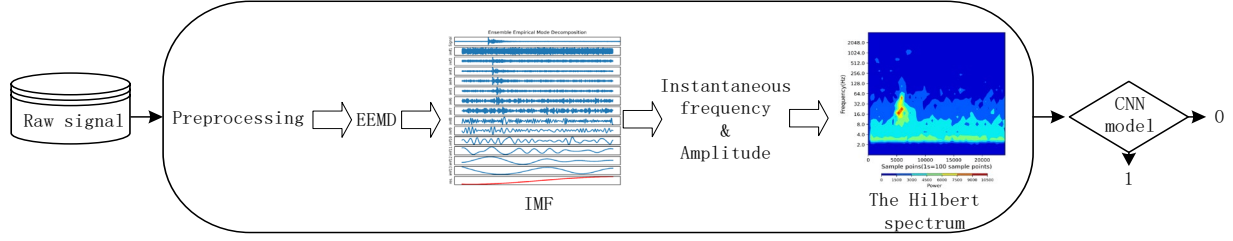


Figure 1. The flow chart of the method in this paper

II. EEMD

Ensemble empirical mode decomposition (EEMD) is an improved method of empirical mode decomposition (EMD), which overcomes the mode mixing problem of EMD. In order to better introduce EEMD, we first introduce EMD. EMD is an adaptive nonlinear non-stationary signal analysis method, it was proposed by [24] in 1998. Its main idea is to separate the various frequency components of the timing signal from the time curve in the form of an intrinsic mode function. Unlike time-frequency analysis methods such as Fourier transform, wavelet transform, and S transform, EMD does not need to assume a basis function to decompose the signal, but adaptively decomposes the signal according to the signal itself. It is considered to be a major breakthrough in linear and steady-state spectrum analysis based on Fourier transform in recent years. In recent years, it has been widely used in marine hydrology, atmospheric monitoring, mechanical fault diagnosis and seismic signal analysis and other fields.

The key to the EMD method is to decompose the signal into a finite number of IMFs. The IMF needs to meet two conditions: (1) The number of extrema and the number of zero-crossings must either be equal or differ at most by one; (2) At any point in time, the average value of the upper envelope defined by the local maximum and the lower envelope defined by the local minimum is 0. The implementation steps of the EMD method are:

1) Find the local maximum and the local minimum of the original signal, and use the cubic spline interpolation to connect the local maximum and local minimum, get the upper envelope $S_{\max}(t)$ and the Lower envelope $S_{\min}(t)$;

2) Calculate the average of the upper envelope and the lower envelope to get the mean envelope $S_{\text{avg}}(t)$, the formula is:

$$S_{\text{avg}}(t) = \frac{1}{2}[S_{\max}(t) + S_{\min}(t)], \quad (1)$$

3) Use original signal $S(t)$ subtract the mean envelope $S_{\text{avg}}(t)$, get the first component, the formula is:

$$h(t) = S(t) - S_{\text{avg}}(t), \quad (2)$$

Check $h(t)$ whether satisfies the two conditions of IMF. If satisfies, consider $h(t)$ as the first IMF component, denote as $c_1(t)$; If the conditions of the IMF are not satisfies, this component is regarded as the original signal $S(t)$; Then

repeat the above steps until $h(t)$ satisfies the two conditions of IMF, at this time the first IMF component of the original signal is obtained. Subtract the first IMF component from the original signal to obtain the remaining term after the first decomposition, the formula is:

$$r_1(t) = S(t) - c_1(t), \quad (3)$$

Generally, the first IMF component represents the highest frequency part of the original signal, and the frequency of the second IMF component is lower than that of the first IMF component.

Since the remaining term $r_1(t)$ still contains a lot of information, it is regarded as a new original signal and the above steps are repeated continuously until the remaining term becomes a monotonic function or constant and no more information can be extracted from it, and the decomposition ends. At this time, we have obtained n IMF components from the highest frequency to the lowest frequency and a trend term or constant term decomposed from the original signal. The original signal can be expressed as the sum of n IMF components and a trend term or constant term. The formula is:

$$S(t) = \sum_{i=1}^n c_i(t) + r_n(t) \quad (4)$$

Although EMD is proven to be an effective time-frequency analysis method, it still has some shortcomings, that is, the phenomenon of mode mixing, which is defined as the existence of characteristic components of different scales in the same IMF component or the same scale is distributed in different IMF points, it will falsely imply that there are different physical processes in one mode.

In order to solve this problem, in the work of [25] proposed a noise-assisted data analysis method EEMD. Its main idea is to add a white noise sequence of finite amplitude during each decomposition process, so that white noise uniformly fills the entire time-frequency space. In the decomposition process, components of different scales are automatically separated, and signals of different scales are adjusted according to the appropriate intrinsic modal function determined by the dyadic filter. EEMD automatically eliminates mode mixing in all cases, which represents the EMD method a major improvement. The implementation steps of the EEMD method are:

1) Add a white noise of finite amplitude $w_i(t)$ to the original signal $S(t)$ to obtain a new signal $S_i(t)$ with white noise added, the formula is:

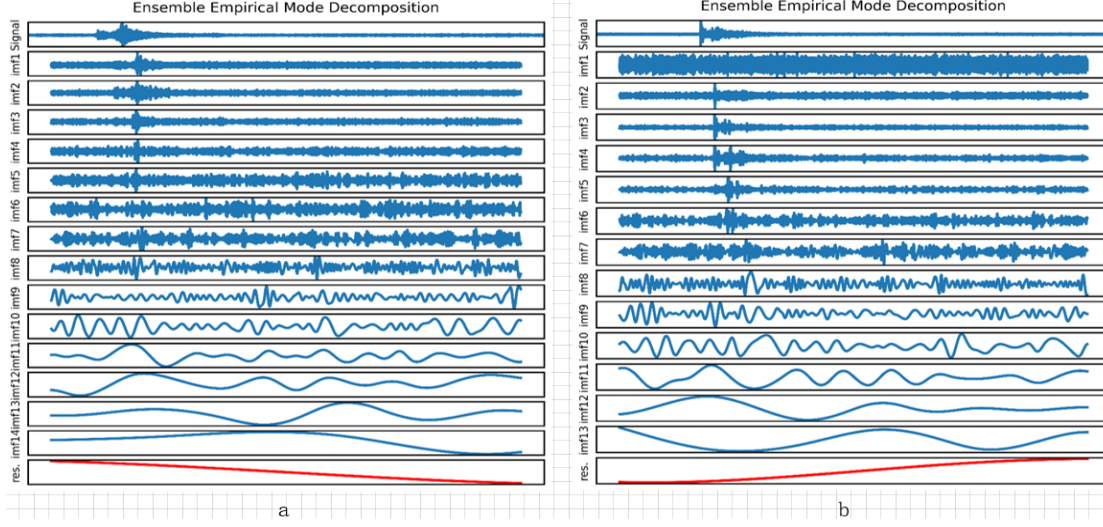


Figure 2. a: IMF component image obtained by decomposing natural seismic signal using EEMD. b: IMF component image obtained by decomposing explosion signal using EEMD.

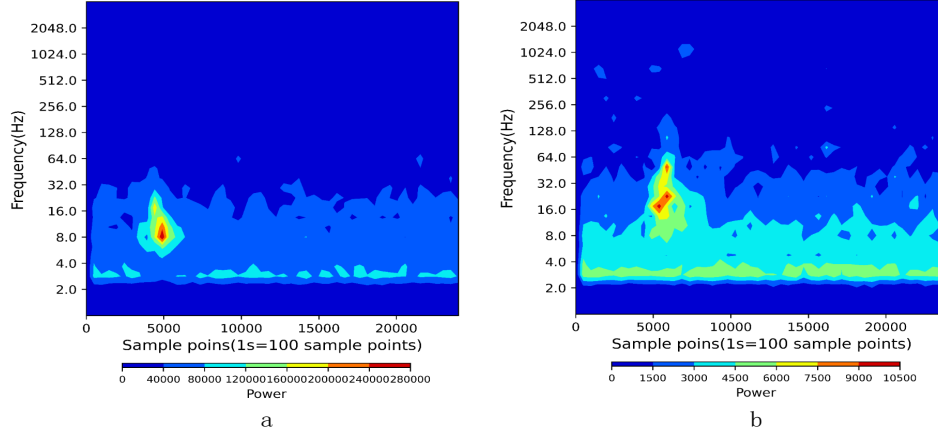


Figure 3. a: the Hilbert spectrum of natural seismic signal. b: the Hilbert spectrum of explosion signal.

$$S_i(t) = S(t) + w_i(t), \quad (5)$$

2) Like the EMD method, decompose the white noise-added data into IMF;

3) Repeat the above steps continuously, and use different white noise for each decomposition;

4) Obtain the mean value of the corresponding IMF as the final decomposition result.

Figure 2 shows the IMF pictures obtained by EEMD of earthquake and explosion signals, respectively.

III. HILBERT SPECTRUM

After completing the EEMD and obtaining the corresponding IMF, we perform the Hilbert transform on each IMF component, the purpose is to obtain the instantaneous frequency energy with actual physical meaning in the signal, and then achieve high-resolution time-frequency analysis, the formula is :

$$Y(t) = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{c(\tau)}{t-\tau} d\tau, \quad (6)$$

in the formula, P represents the Cauchy principal value. Construct its analytical signal:

$$z_i(t) = c_i(t) + iY(t) = a_i(t)e^{i\theta_i(t)}, \quad (7)$$

the amplitude and phase functions are respectively:

$$a_i(t) = \sqrt{c_i^2(t) + Y_i^2(t)}, \quad (8)$$

$$\theta_i(t) = \arctan \frac{Y_i(t)}{c_i(t)}, \quad (9)$$

further, the instantaneous frequency can be calculated, the formula is:

$$\omega_i(t) = \frac{d\theta_i(t)}{dt}, \quad (10)$$

according to the amplitude function and instantaneous frequency, the original signal can be expressed by the following formula:

$$S(t) = \sum_{i=1}^n a_i(t)e^{i \int \omega_i(t) dt}, \quad (11)$$

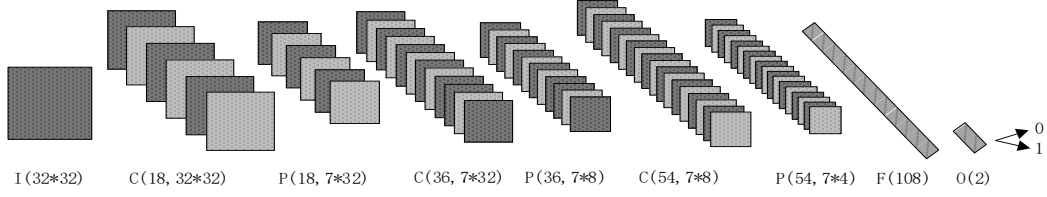


Figure 4. The structure of the convolutional neural network used in the method in this paper. I, C, P, F, O represent input, convolutional layer, pooling layer, fully connected layer and output layer, respectively.

after the Hilbert transform is performed on each IMF component, a three-dimensional Hilbert spectrum of time, frequency and amplitude can be constructed according to the instantaneous frequency and amplitude. The local amplitude and instantaneous frequency obtained from the intrinsic mode function through the Hilbert transform can give us a complete data energy-frequency-time distribution. Figure 3 shows the Hilbert spectrum of the natural earthquake and artificial blast signal IMF.

IV. CNN MODEL

In our network model, we refer to the classic convolutional neural network model LeNet5. Our neural network structure mainly includes three convolutional layers, three maximum pooling layers, and two fully connected layers. After the connection layer, we use the ReLU activation function to perform activation operations. We also add a dropout layer before the network output layer to prevent overfitting. Among them, the convolution kernel of the first convolution layer is 13×9 , the input channel is 1, the output channel is 18, the convolution kernel of the second convolution layer is 3×7 , the input channel is 18, and the output channel is 36, the convolution kernel of the third convolution layer is 3×3 , the input channel is 36, and the output channel is 54; the first pooling layer is the maximum pooling of 5×1 , and the second pooling layer is the maximum pooling of 1×4 , the third pooling layer is the maximum pooling of 1×2 , and the number of neurons in the fully connected layer is 108. The loss function of the network we use is the cross-entropy loss function, and the optimization algorithm uses the Adam algorithm. Figure 4 shows the structure of the convolutional neural network used in the method in this paper.

V. EXPERIMENTS SETUP

A. Data

In our work, the data used are two types of seismic signals: natural earthquake and artificial explosion. There are 54 natural seismic events and 63 explosion events. We regard a channel waveform of an event as an identification unit. After manual processing by seismic experts, the effective waveforms obtained from earthquakes and artificial blasts were 1,674 and 1,509, respectively. We divide the two types of data into training set and test set. The training set is 80% and the test set is 20%. The training set is used for convolutional neural network model training to find the optimal network model. The test set is used for model testing.

B. Experimental Details

In the experiment, we first perform some necessary preprocessing on the original seismic signal, detrend the data, intercept the signal to a uniform length, and then use the EEMD method mentioned in Section 1 to perform the preprocessing on the preprocessed signal. Decompose to obtain the corresponding IMF component, and then use the formula (8) and formula (10) mentioned in Section 2 to find the instantaneous frequency and amplitude of the corresponding IMF component, and use the Hilbert transform of formula (6) to these IMF components are transformed to generate a Hilbert spectrum as the input of the convolutional neural network.

Since the size of the original Hilbert spectrum generated is very large, the length and width are about 3800 pixels, that is, the parameter amount of a sample can reach more than 14,440,000, which is a very huge parameter amount for the neural network model, and computer resource consumption and time cost are also very huge. Since our neural network model refers to the LeNet5 model, we also refer to the input data of LeNet5 in the input of the neural network. In our experiments, we reset the original size of the Hilbert spectrum to 16×16 , 32×32 , 48×48 , and 64×64 pixels, and extracted their three-channel image features and gray levels image features respectively, as the input of the neural network, after our experimental comparison, the 32×32 pixel gray image features perform best in the classification model, so we use the 32×32 pixel gray image features as the final neural network input of.

C. Result

We used 80% of the data set for model training. The natural seismic signals and explosion signals used to train the model were 1,339 and 1,207, respectively. The remaining 20% were used for model testing, natural seismic signals and explosion signals, there are 335 and 302, respectively. After completing the training of the network model, we used the five-fold cross-validation method to test the model, and the highest classification accuracy rate was 94.98%, and the average classification accuracy rate was 94.41%. We also used the same data to conduct comparative experiments on the SVM method and the MLP method. The highest classification accuracy rate of the SVM method was 89.17%, the average classification accuracy rate was 87.94%, and the highest classification accuracy rate of the MLP method obtained was 86.97. %, the average classification accuracy rate is 85.65%, and the specific test results are shown in Table 1.

TABLE I. TEST RESULTS OF MULTI-LAYER PERCEPTRON(MLP), SUPPORT VECTOR MACHINE(SVM) AND THE METHOD IN THIS PAPER IN THE FIVE-FOLD CROSS-VALIDATION METHOD

	Test 1(%)	Test 2(%)	Test 3(%)	Test 4(%)	Test 5(%)	Avg(%)
MLP	85.71	86.97	86.19	84.77	84.61	85.65
SVM	87.44	87.28	89.01	89.17	86.81	87.94
Propose method	94.35	94.19	94.03	94.51	94.98	94.41

VI. CONCLUSION AND DISCUSSION

The experimental results show that using the EEMD method to decompose the original signal can adaptively decompose the IMF components of the different frequency components of the signal according to the signal itself, and obtain a set of IMF classifications from high to low frequencies, and then using the Hilbert transform to transformed the extracted IMF components, obtain a high-resolution three-dimensional Hilbert spectrum of time, frequency and amplitude. The Hilbert spectrum is used as the input of the convolutional neural network. After model training, the test accuracy can reach up to 94.98%. The average accuracy rate is 94.41%, which shows the effectiveness of this method in the classification of earthquakes and explosion seismic signals.

The classification of seismic signals is of great significance for updating clear earthquake catalog, early-warning the arrival of strong earthquake. With the improvement of the accuracy of seismic station instruments, more and more seismic signals will be collected and recorded. This will bring great pressure to our staff. Therefore, The development of an automatic classification system for seismic signals will greatly reduce the pressure on the staff and at the same time bring positive significance to the research in the field of seismology.

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